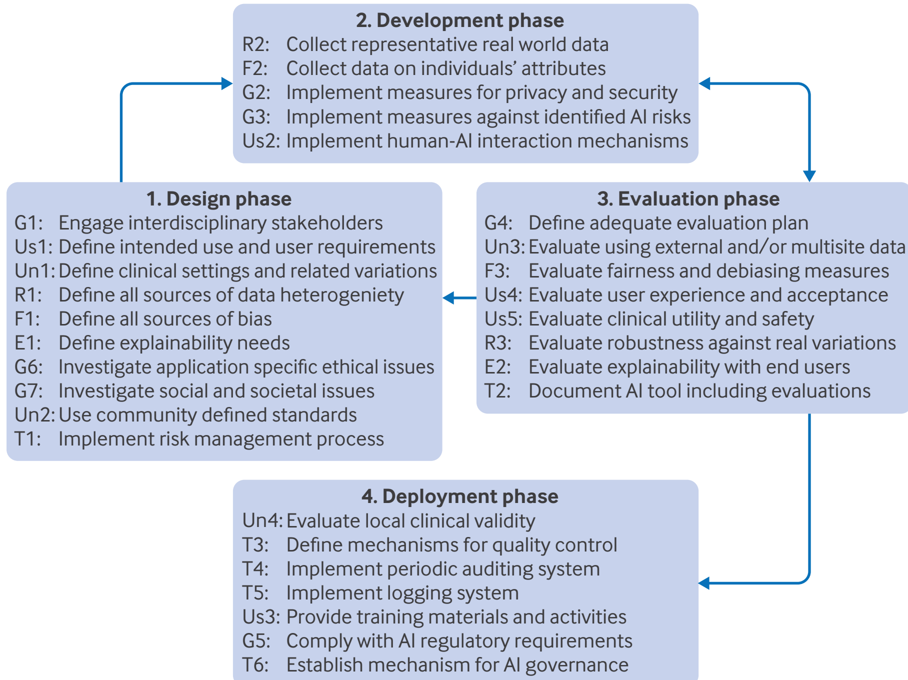

Fig 3 | Embedding the FUTURE-AI best practices into an agile process throughout the artificial intelligence (AI) lifecycle. E=explainability; F=fairness; G=general; R=robustness; T=traceability; Un=universality; Us=usability

- Finally, the deployment phase is dedicated to ensuring local validity, providing training, implementing monitoring mechanisms, and ensuring regulatory compliance for adoption in real world healthcare practice.

Operationalisation of FUTURE-AI

In this section, we provide a detailed list of practical steps for each recommendation, accompanied by specific examples of approaches and methods that can be applied to operationalise each step towards

Table 3 | Practical steps and examples to implement FUTURE-AI recommendations during design phase

Recommendations	Operations	Examples
Engage interdisciplinary stakeholders (general 1)	Identify all relevant stakeholders	Patients, GPs, nurses, ethicists, data managers[82 83]
	Provide information on the AI tool and AI	Educational seminars, training materials, webinars[84]
	Set up communication channels with stakeholders	Regular group meetings, one-to-one interviews, virtual platform[85]
	Organise cocreation consensus meetings	One day cocreation workshop with n=15 multidisciplinary stakeholders[86]
	Use qualitative methods to gather feedback	Online surveys, focus groups, narrative interviews[87]
Define intended use and user requirements (usability 1)	Define the clinical need and AI tool's goal	Risk prediction, disease detection, image quantification
	Define the AI tool's end users	Patients, cardiologists, radiologists, nurses
	Define the AI model's inputs	Symptoms, heart rate, blood pressure, ECG, image scan, genetic test
	Define the AI tool's functionalities and interfaces	Data upload, AI prediction, AI explainability, uncertainty estimation[88]
	Define requirements for human oversight	Visual quality control, manual corrections[89 90]
	Adjust user requirements for all end user subgroups	According to role, age group, digital literacy level[91]
Define intended clinical settings and cross setting variations (universality 1)	Define the AI tool's healthcare setting(s)	Primary care, hospital, remote care facility, home care
	Define the resources needed at each setting	Personnel (experience, digital literacy), medical equipment (eg, >1.5 T MRI scanner), IT infrastructure
	Specify if the AI tool is intended for high end and/or low resource settings	Facilities with MRI scanners >1.5 T *v* low field MRIs (eg, 0.5 T), high end *v* low cost portable ultrasound[92 93]
	Identify all cross settings variations	Data formats, medical equipment, data protocols, IT infrastructure[94]
Define sources of data heterogeneity (robustness 1)	Engage relevant stakeholders to assess data heterogeneity	Clinicians, technicians, data managers, IT managers, radiologists, device vendors
	Identify equipment related data variations	Differences in medical devices, manufacturers, calibrations, machine ranges (from low cost to high end)[95]
	Identify protocol related data variations	Differences in image sequences, data acquisition protocols,[96] data annotation methods, sampling rates, preprocessing standards
	Identify operator related data variations	Different in experience and proficiency, operator fatigue, subjective judgment, technique variability
	Identify sources of artefacts and noises	Image noise, motion artefacts, signal dropout, sensor malfunction
	Identify context specific data variations	Lower data quality acquisition in emergency units, during high patient volume times

(Continued)

Table 3 | Continued

Recommendations	Operations	Examples
Define any potential sources of bias (fairness 1)	Engage relevant stakeholders to define the sources of bias	Patients, clinicians, epidemiologists, ethicists, social carers[97 98]
	Define standard attributes that might affect the AI tool's fairness	Sex, age, socioeconomic status[99]
	Identify application specific sources of bias beyond standard attributes	Skin colour for skin cancer detection,[100 101] breast density for breast cancer detection[34]
	Identify all possible human biases	Data labelling, data curation[99]
Define the need and requirements for explainability with end users (explainability 1)	Engage end users to define explainability requirements	Clinicians, technicians, patients[102]
	Specify if explainability is necessary	Not necessary for AI enabled image segmentation part, critical for AI enabled diagnosis
	Specify the objectives of AI explainability (if it is needed)	Understanding AI model, aiding diagnostic reasoning, justifying treatment recommendations[103]
	Define suitable explainability approaches	Visual explanations, feature importance, counterfactuals[104]
	Adjust the design of the AI explanations for all end user subgroup	Heatmaps for clinicians, feature importance for patients[105 106]
Investigate ethical issues (general 6)	Consult ethicists on ethical considerations	Ethicists specialised in medical AI and/or in the application domain (eg, paediatrics)[107]
	Assess if the AI tool's design is aligned with relevant ethical values	Right to autonomy, information, consent, confidentiality, equity[107]
	Identify application specific ethical issues	Ethical risks for a paediatric AI tool (eg, emotional impact on children)[108 109]
	Comply with local ethical AI frameworks	AI ethical guidelines from Europe,[2] United Kingdom,[110 111] United Sates,[112] Canada,[113] China,[114] India,[115] Japan,[116 117] Australia,[118] etc
Investigate social and environmental issues (general 7)	Investigate AI tool's social and environmental impact	Workforce displacement, worsened working conditions and relations, deskilling,[80] dehumanisation of care, reduced health literacy, increased carbon footprint,[119] negative public perception[107 120]
	Define mitigations to enhance the AI tool's social and environmental impact	Interfaces for physician-patient communication, workforce training, educational programmes, energy efficient computing practices, public engagement initiatives
	Optimise algorithms, energy efficiency	Develop and use energy efficient algorithms that minimise computational demands. Techniques like model pruning, quantisation, and edge computing can reduce the energy required for AI tasks
	Promote responsible data usage	Focus on collecting and processing only the necessary amount of data. Implement federated learning techniques to minimise data transfers. This approach keeps data localised, reducing need for extensive data movement, which consumes energy
	Monitor and report the environmental impact of the AI tool	Regularly monitor and report on the environmental impact of AI systems used in healthcare, including energy usage, carbon emissions, and waste generation
Use community defined standards (universality 2)	Use a standard definition for the clinical task	Definition of heart failure by the American Academy of Cardiology[121]
	Use a standard method for data labelling	BI-RADS for breast imaging[122]
	Use a standard ontology for the AI inputs	DICOM for imaging data,[123] SNOMED for clinical data[41]
	Adopt technical standards	IEEE 2801-2022 for medical AI software[43]
	Use standard evaluation criteria	See Maier-Hein et al[21] for medical imaging applications, Barocas et al[31] and Bellamy et al[35] for fairness evaluation
Implement a risk management process (traceability 1)	Identify all possible clinical, technical, ethical, and societal risks	Bias against under-represented subgroups, limited generalisability to low resource facilities, data drift, lack of acceptance by end users, sensitivity to noisy inputs[124]
	Identify all possible operational risks	Misuse of the AI tool (owing to insufficient training or not following the instructions), application of the AI tool outside of the target population (eg, individuals with implants), use of the tool by others than the target end users (eg, technician instead of physician), hardware failure, incorrect data annotations, adversarial attacks[76 125]
	Assess the likelihood of each risk	Very likely, likely, possible, rare
	Assess the consequences of each risk	Patient harm, discrimination, lack of transparency, loss of autonomy, patient reidentification[126]
	Prioritise all the risks depending on their likelihood and consequences	Risk of bias (if no personal attributes are included in the model) *v* risk of patient reidentification (if personal attributes are collected)
	Define mitigation measures to be applied during AI development	Data enhancement, data augmentation,[127] bias correction techniques, domain adaptation,[66] transfer learning,[128] continuous learning[129]
	Define mitigation measures to be applied after deployment	Warnings to the users, system shutdown, reprocessing of the input data, acquisition of new input data, use of an alternative procedure, or human judgment only
	Set up a mechanism to monitor and manage risks over time	Periodic risk assessment every six months
	Create a comprehensive risk management file	Including all risks, their likelihood and consequences, risk mitigation measures, risk monitoring strategy

AI=artificial intelligence; BI-RADS=breast imaging reporting and data system; DI-COM=Digital Imaging and Communications in Medicine; ECG=electrocardiogram; GP=general practitioner; IEEE=Institute of Electrical and Electronics Engineers; MRI=magnetic resonance imaging; SNOMED=Systematized Nomenclature of Medicine.

trustworthy AI, as shown in table 3, table 4, table 5, and table 6. This approach offers easy-to-use, step-by-step guidance for all end users of the FUTURE-AI framework when designing, developing, validating and deploying new AI tools for healthcare.

Discussion

Despite the tremendous amount of research in medical AI in recent years, currently only a limited number of AI tools have made the transition to clinical practice. Although many studies have shown the huge potential

Table 4 | Practical steps and examples to implement FUTURE-AI recommendations during development phase

Recommendations	Operations	Examples
Collect representative training dataset (robustness 2)	Collect training data that reflect the demographic variations	According to age, sex, ethnicity, socioeconomics
	Collect training data that reflect the clinical variations	Disease subgroups, treatment protocols, clinical outcomes, rare cases
	Collect training data that reflect variations in real world practice	Data acquisition protocols, data annotations, medical equipment, operational variations (eg, patient motion during scanning)[125]
	Artificially enhance the training data to mimic real world conditions	Data augmentation,[127] data synthesis (eg, low quality data, noise addition),[130] data harmonisation,[131 132] data homogenisation[133]
Collect information on individuals' and data attributes (fairness 2)	Request approval for collecting data on personal attributes	Sex, age, ethnicity, socioeconomic status[134]
	Collect information on standard attributes of the individuals (if available and allowed)	Sex, age, nationality, education[135]
	Include application specific information relevant for fairness analysis	Skin colour, breast density,[34] presence of implants, comorbidity[136]
	Estimate data distributions across subgroups	Male *v* female, across ethnic groups
	Collect information on data provenance	Data centres, equipment characteristics, data preprocessing, annotation processes
Implement measures for data privacy and security (general 2)	Implement measures to ensure data privacy and security	Data deidentification, federated learning,[137 138] differential privacy, encryption[139]
	Implement measures against malicious attacks	Firewalls, intrusion detection systems, regular security audits[139]
	Adhere to applicable data protection regulations	General Data Protection Regulation,[140] Health Insurance Portability and Accountability Act[141]
	Define suitable data governance mechanisms	Access control, logging system
Implement measures against identified AI risks (general 3)	Implement a baseline AI model and identify its limitations	Bias, lack of generalisability[142]
	Implement methods to enhance robustness to real world variations (if needed)	Regularisation,[143] data augmentation,[127] data harmonisation,[131] domain adaptation[66]
	Implement methods to enhance generalisability across settings (if needed)	Regularisation, transfer learning,[144] knowledge distillation[145]
	Implement methods to enhance fairness across subgroups (if needed)	Data resampling, bias free representation,[36] equalised odds postprocessing[37 38 146]
Establish mechanisms for human-AI interactions (usability 2)	Implement mechanisms to standardise data preprocessing and labelling	Data preprocessing pipeline, data labelling plugin
	Implement an interface for using the AI model	Application programming interface
	Implement interfaces for explainable AI	Visual explanations, heatmaps, feature importance bars[105 106]
	Implement mechanisms for user centred quality control of the AI results	Visual quality control, uncertainty estimation[147]
	Implement mechanism for user feedback	Feedback interface[148]

AI=artificial intelligence.

of AI to improve healthcare, major clinical, technical, socioethical, and legal challenges persist.

In this paper, we presented the results of an international effort to establish a consensus guideline for developing trustworthy and deployable AI tools in healthcare. To this end, the FUTURE-AI Consortium was established, which provided knowledge and expertise across a wide range of disciplines and stakeholders, resulting in consensus and wide support, both geographically and across domains. Through an iterative process that lasted 24 months, the FUTURE-AI framework was established, comprising a comprehensive and self-contained set of 30 recommendations, which covers the whole lifecycle of medical AI. By dividing the recommendations across six guiding principles, the pathways towards responsible and trustworthy AI are clearly characterised. Because of its broad coverage, the FUTURE-AI guideline can benefit a wide range of stakeholders in healthcare, as detailed in table 2 in the appendix.

FUTURE-AI is a risk informed framework, proposing to assess application specific risks and challenges early in the process (eg, risk of discrimination, lack of generalisability, data drifts over time, lack of acceptance by end users, potential harm for patients, lack of transparency, data security vulnerabilities, ethical risks), followed by implementing tailored measures to reduce these risks (eg, collect data on individuals' attributes to assess and mitigate bias). As the specific measures to be implemented have benefits and potential weaknesses that the developers need to assess and take into consideration, a risk-benefit balancing trade-off has to be made. For example, collecting data on individuals' attributes might increase the risk of reidentification, but can enable the risk of bias and discrimination to be reduced. Therefore, in FUTURE-AI, risk management (as recommended in traceability 1) must be a continuous and transparent process throughout the AI tool's lifecycle.

FUTURE-AI is also an assumption-free, highly collaborative framework, recommending to continuously engage with multidisciplinary stakeholders to understand application specific needs, risks, and solutions (general 1). This is crucial to investigate all possible risks and factors that might reduce trust in a specific AI tool. For example, instead of making any assumptions on possible sources of bias, FUTURE-AI recommends that AI developers engage with healthcare professionals, domain experts, representative citizens, and/or ethicists early in the process to form interdisciplinary AI development teams and investigate in-depth the application specific

Table 5 | Practical steps and examples to implement FUTURE-AI recommendations during evaluation phase

Recommendations	Operations	Examples
Define adequate evaluation plan (general 4)	Identify the dimensions of trustworthy AI to be evaluated	Robustness, clinical safety, fairness, data drifts, usability, explainability
	Select appropriate testing datasets	External dataset from a new hospital, public benchmarking dataset[148]
	Compare the AI tool against standard of care	Conventional risk predictors, visual assessment by radiologist, decision by clinician[149 150]
	Select adequate evaluation metrics	F1 score for classification, concordance index for survival,[21] statistical parity for fairness[151]
Evaluate using external datasets and/or multiple sites (universality 3)	Identify relevant public datasets	Cancer Imaging Archive,[152] UK Biobank,[153] M&Ms,[154] MAMA-MIA,[155] BRATS[156]
	Identify external private datasets	New prospective dataset from same site or from different clinical centre[157 158]
	Select multiple evaluation sites	Three sites in same country, five sites in two different countries
	Verify that evaluation data and sites reflect real world variations	Variations in demographics, clinicians, equipment
	Confirm that no evaluation data were used during training	Yes/no
Evaluate fairness and bias correction measures (fairness 3)	Select attributes and factors for fairness evaluation	Sex, age, skin colour, comorbidity
	Define fairness metrics and criteria	Statistical parity difference defined fairness between −0.1 and 0.1[35]
	Evaluate fairness and identify biases	Fair with respect to age, biased with respect to sex
	Evaluate bias mitigation measures	Training data resampling,[159] equalised odds postprocessing[37 38 146]
	Evaluate impact of mitigation measures on model performance	Data resampling removed sex bias but reduced model performance[160]
	Report identified and uncorrected biases	In AI information leaflet and technical documentation[161] (see traceability 2).
Evaluate user experience (usability 4)	Evaluate usability with diverse end users	According to sex, age, digital proficiency level, role, clinical profile[162 163]
	Evaluate user satisfaction using usability questionnaires	System usability scale[164]
	Evaluate user performance and productivity	Diagnosis time with and without AI tool, image quantification time[165]
	Assess training of new end users	Average time to reach competency, training difficulties[166]
Evaluate clinical utility and safety (usability 5)	Define clinical evaluation plan	Randomised control trial,[59 167] in silico trial[168]
	Evaluate if AI tool improves patient outcomes	Better risk prevention, earlier diagnosis, more personalised treatment[169]
	Evaluate if AI tool enhances productivity or quality of care	Enhanced patient triage, shorter waiting times, faster diagnosis, higher patient intake[169]
	Evaluate if AI tool results in cost savings	Reduction in diagnosis costs,[170 171] reduction in overtreatment[172]
	Evaluate AI tool's safety	Side effects or major adverse events in randomised control trials[173 174]
Evaluate robustness (robustness 3)	Evaluate robustness under real world variations	Using test-retest datasets,[175 176] multivendor datasets[177]
	Evaluate robustness under simulated variations	Using simulated repeatability tests,[148] synthetic noise and artefacts (eg, image blurring)[178]
	Evaluate robustness against variations in end users	Different technicians or annotators
	Evaluate mitigation measures for robustness enhancement	Regularisation,[63] data augmentation,[64 127] noise addition, normalisation,[179] resampling, domain adaptation[66]
Evaluate explainability (explainability 2)	Assess if explanations are clinically meaningful	Reviewing by expert panels, alignment to current clinical guidelines, explanations not pointing to shortcuts[73]
	Assess explainability quantitatively using objective measures	Fidelity, consistency, completeness, sensitivity to noise[180-182]
	Assess explainability qualitatively with end users	Using user tests or questionnaires to measure confidence and affect clinical decision making[183 184]
	Evaluate if explanations cause end user overconfidence or overreliance	Measure changes in clinician confidence,[185 186] performance with and without AI tool[187]
	Evaluate if explanations are sensitive to input data variations	Stress tests under perturbations to evaluate the stability of explanations[74 188]
Provide documentation (traceability 2)	Report evaluation results in publication using AI reporting guidelines	Peer reviewed scientific publication using TRIPOD-AI reporting guideline[15]
	Create technical documentation for AI tool	AI passport,[189] model cards[49] (including model hyperparameters, training and testing data, evaluations, limitations, etc)
	Create clinical documentation for AI tool	Guidelines for clinical use, AI information leaflet (including intended use, conditions and diseases, targeted populations, instructions, potential benefits, contraindications)
	Provide risk management file	Including identified risks, mitigation measures, monitoring measures
	Create user and training documentation	User manuals, training materials, troubleshooting, FAQs (see usability 2)
	Identify and provide all locally required documentation	Compliance documents and certifications (see general 5)

AI=artificial intelligence; FAQ=frequently asked questions.

sources of bias, which might include domain specific attributes (eg, breast density for AI applications in breast cancer).

The FUTURE-AI guideline was defined in a generic manner to ensure it can be applied across a variety of domains (eg, radiology, genomics, mobile health, electronic health records). However, for many recommendations, their applicability varies across medical use cases, even within domains. To this end, the first recommendation in each of the guiding principles is to identify the specificities to be addressed, such as the types of biases (fairness 1), the clinical settings (universality 1), or the need and approaches for explainable AI (explainability 1). This facilitates generalisability across domains, but also ensures sustainability for future use. Furthermore, we recognise that a one-size-fits-all approach is not feasible, as addressal of many of the recommendations is use case specific, and standards do not exist yet or are subject to change. Therefore, we focused on developing best practices for enhancing the trustworthiness of medical AI tools, while consciously avoiding the imposition of specific techniques for the implementation of each recommendation. This flexibility also acknowledges the diversity of methods for tackling challenges and mitigating risks

Table 6 | Practical steps and examples to implement FUTURE-AI recommendations during deployment phase

Recommendations	Operations	Examples
Evaluate and demonstrate local clinical validity (universality 4)	Test AI model using local data	Data from local clinical registry
	Identify factors that could affect AI tool's local validity	Local operators, equipment, clinical workflows, acquisition protocols
	Assess AI tool's integration within local clinical workflows	AI tool's interface aligns with hospital IT system[148] or disrupts routine practice
	Assess AI tool's local practical utility and identify any operational challenges	Time to operate, clinician satisfaction, disruption of existing operations[148 190]
	Implement adjustments for local validity	Model calibration, fine-tuning,[191] transfer learning[192-194]
	Compare performance of AI tool with that of local clinicians	Side-by-side comparison, in silico trial
Define mechanisms for quality control of AI inputs and outputs (traceability 3)	Implement mechanisms to identify erroneous input data	Missing value or out-of-distribution detector,[195] automated image quality assessment[73 196 197]
	Implement mechanisms to detect implausible AI outputs	Postprocessing sanity checks, anomaly detection algorithm[198]
	Provide calibrated uncertainty estimates to inform on AI tool's confidence	Calibrated uncertainty estimates per patient or data point[53 54 199]
	Implement system for continuous quality monitoring	Real time dashboard tracking data quality and performance metrics[200]
	Implement feedback mechanism for users to report issues	Feedback portal enabling clinicians to report discrepancies or anomalies
Implement system for periodic auditing and updating (traceability 4)	Define schedule for periodic audits	Biannual or annual
	Define audit criteria and metrics	Accuracy, consistency, fairness, data security[148]
	Define datasets for periodic audits	Newly acquired prospective dataset from local hospital
	Implement mechanisms to detect data or concept drifts	Detecting shifts in input data distributions[148 190]
	Assign role of auditor(s) for AI tool	Internal auditing team, third party company[190]
	Update AI tool based on audit results	Updating AI model,[56] re-evaluating AI model,[148] adjusting operational protocols, continuous learning[201-204]
	Implement reporting system from audits and subsequent updates	Automatic sharing of detailed reports to healthcare managers and clinicians
	Monitor impact of AI updates	Impact on system performance and user satisfaction[56]
Implement logging system for usage recording (traceability 5)	Implement logging framework capturing all interactions	User actions, AI inputs, AI outputs, clinical decisions
	Define data to be logged	Timestamp, user ID, patient ID (anonymised), action details, results
	Implement mechanisms for data capture	Software to automatically record every data and operation
	Implement mechanisms for data security	Encrypted log files, privacy preserving techniques[205]
	Provide access to logs for auditing and troubleshooting	By defining authorised personnel, eg, healthcare or IT managers
	Implement mechanism for end users to log any issues	A user interface to enter information about operational anomalies
	Implement log analysis	Time series statistics and visualisations to detect unusual activities and alert administrators
Provide training (usability 3)	Create user manuals	User instructions, capabilities, limitations, troubleshooting steps, examples, and case studies
	Develop training materials and activities	Online courses, workshops, hands-on sessions
	Use formats and languages accessible to intended end users	Multiple formats (text, video, audio) and languages (English, Chinese, Swahili)
	Customise training to all end user groups	Role specific modules for specialists, nurses, and patients
	Include training to enhance AI and health literacy	On application specific AI concepts (eg, radiomics, explainability), AI driven clinical decision making
Identify and comply with applicable AI regulatory requirements (general 5)	Engage regulatory experts to investigate regulatory requirements	Regulatory consultants from intended local settings
	Identify specific regulations based on AI tool's intended markets	FDA's SaMD in the United States,[206] MDR and AI Act[207] in the EU
	Identify specific requirements based on AI tool's purpose	De Novo classification (Class III)[208]
	Define list of milestones towards regulatory compliance	MDR certification: technical verification, pivotal clinical trial, risk and quality management, postmarket follow-up
Establish mechanisms for AI governance (traceability 6)	Assign roles for AI tool's governance	For periodic auditing, maintenance, supervision (eg, healthcare manager)
	Define responsibilities for AI related errors	Responsibilities of clinicians, healthcare centres, AI developers, and manufacturers
	Define mechanisms for accountability	Individual *v* collective accountability/liability,[25] compensations, support for patients

AI=artificial intelligence; FDA=United States Food and Drug Administration; MDR=medical device regulation; SaMD= Software as a Medical Device.

in medical AI. For example, the recommendation to protect personal data during AI training can be implemented through data deidentification, federated learning, differential privacy or encryption, among other methods. While such concrete examples are listed in this article, especially in table 3, table 4, table 5, and table 6, the most adequate techniques for implementing each recommendation should be ultimately selected by the AI development team as a function of the application domain, clinical use case, and data characteristics, as well as the advantages and limitations of each method. Similarly, all stakeholders of the AI development team are together responsible for addressing the recommendations, where the role of each party might vary per application, method, domain, project setup, and use case.

While the FUTURE-AI framework offers insights for regulating medical AI, future work is needed to incorporate these recommendations into regulatory procedures. For example, we propose mechanisms to enhance traceability and governance, such as logging. However, the crucial issue of liability is yet